THE NATURAL PERMANENT MAGNETIZATION OF THE LOWER BASALTS OF NORTHERN IRELAND

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Summary

On cooling, lava flows become permanently magnetized in the direction of the local geomagnetic field, and studies of rock magnetism therefore allow the direction of the geomagnetic field in the past to be traced. In this paper the natural permanent magnetization of the so-called Lower Basalts of Northern Ireland, which are probably of Eocene age, is investigated. It is found that all lava flows studied possess reverse magnetization, i.e. a magnetization approximately opposite to that of the present field.

The mean direction of magnetization of geologically speaking recent lava flows in Iceland, covering a period of several thousands of years, agrees closely with the direction of the theoretical dipole field (i.e. the field due to a geocentric axial dipole). This leads to the conclusion that, taken over a period of several thousands of years, the magnetic poles centre on the geographic poles. In the case of the Tertiary Northern Ireland lava flows it is found that the mean direction of magnetization is slightly different from the direction of the present dipole field but radically different from the direction of the dipole field calculated for the position of the geographic north pole in Eocene times as proposed by advocates of the polar wandering hypothesis. Though the former difference may conceivably be due to local geological processes, it may represent a real shift of the geographical pole. If the age of the Lower Basalts is Eocene this shift is much less than has been suggested on the polar wandering hypothesis, though this would not be so if the lavas are considerably younger.

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1. *Introduction.*—It has been known for a long time, in fact since about A.D. 1800, that eruptive rocks have a permanent magnetization which is much larger than can be acquired in the present geomagnetic field at 20 deg. C. Experimentally it has been established that the permanent (thermo-remanent) magnetization acquired by an igneous rock which has been heated and cooled through its Curie point in a magnetic field of the order of the Earth's is considerably greater than the permanent magnetization acquired by exposing the specimen to the same field at room temperature. The direction of the thermo-remanent magnetization is the same as that of the ambient field (*cf.* 1). It is therefore likely that the natural direction of magnetization of an igneous rock is the same as that of the local geomagnetic field at the time of its solidification, allowing for subsequent tectonic movements.

This opinion is supported by measurements of the direction of magnetization of historic lava flows. Chevallier (2) determined the direction of the permanent magnetization of dated lava flows on Mt. Etna, and was able to trace the changes in direction from the 12th to the 17th century. The later results agree with

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observatory determinations. Minakami (3) reports that the Yoridai Sawa lava flow (erupted on 1940 July 12) actually became magnetized in the direction of the local geomagnetic field. Hospers (4) found that the same is true of the historic lava flows of Mt. Hekla, Iceland, ranging in age from A.D. 1766 to A.D. 1947–8.

It therefore seems reasonable to obtain data on the history of the geomagnetic field from measurements of the direction of the permanent magnetization of igneous rocks. In this fashion Hospers (4) has studied the history of the geomagnetic field in Iceland from Miocene times to the present day (that is, during the last 20×10^6 years). These investigations have shown that the natural permanent magnetization of the Icelandic lava flows is of thermo-remanent origin, i.e. it originated when these flows cooled down after their extrusion. This magnetization was acquired in the direction of the local geomagnetic field at that time and it has been stable in direction over the period of time that has elapsed since these flows consolidated. For the oldest lava flows of Iceland which have been investigated, this amounts to 20×10^6 years.

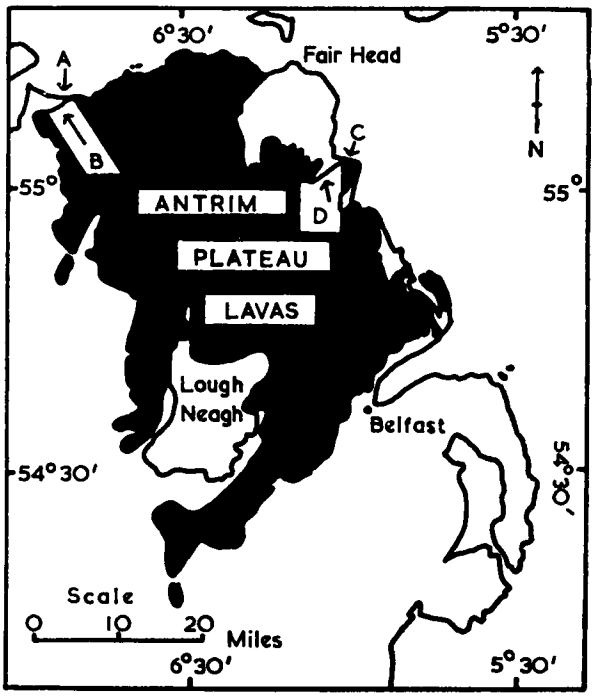


FIG. 1.—*Sketch map of north-east Ireland, showing the extent of the Tertiary plateau basalts and the localities where samples were taken. A=Downhill Railway Station, B=Woodtown, C=Evish, D=section on the southern side of the Glenariff River. After Charlesworth* (5, *Fig.* 39, *p.* 109).

A somewhat similar investigation is presented here of the Lower Basalts of Northern Ireland. In Northern Ireland (Fig. 1) a pile of Tertiary plateau basalts occurs. The flows which make up this series are dominantly basaltic and were extruded sub-aerially. This pile is divided into two by the so-called inter-basaltic horizon which represents a period of volcanic quiescence and separates the Lower Basalts below from the Upper Basalts above (*cf.* 5, p. 142). These Lower Basalts were sampled at four localities. The age of the Lower

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Basalts is generally considered (*cf.* **5**, p. 144; **6**, p. 437; **7**, p. 41) to be Eocene (but *cf.* below).

2. *Sampling.*—Oriented samples were collected at four different localities (*cf.* Fig. 1 and **8**):

A. Near Downhill Railway Station. Geographical coordinates of this section: 6° 49′·3 W., 55° 9′·9 N. In this section 6 lava flows occur, all of which were sampled. The average thickness of the flows is 23 ft.

B. Near Woodtown. (6° 52′·5 W., 55° 9′·2 N.) In this section 9 flows occur, of which 7 were sampled. The average thickness is 23 ft.

C. At Evish, ¾ mile east of Milltown. (6° 1′·4 W., 55° 3′·0 N.) In this section 8 flows occur, of which 6 were sampled. The average thickness is 24 ft.

D. On the southern side of the Glenariff River, near Craignagat. (6° 4′·4 W., 55° 1′·4 N.) Here 7 flows are exposed, of which 5 were sampled. The average thickness is 23 ft.

Samples of known orientation were obtained by drawing on the sample *in situ* a horizontal line on at least two faces and an arrow of known bearing on the top. This bearing was found by drawing the arrow towards a distant and identifiable land-mark, the bearing of which was subsequently read from the map. Magnetic bearings were avoided, as the magnetic declination varies greatly near the edge of the plateau basalt series. Specimens suitable for accurate measurement of the direction of permanent magnetization were prepared in the laboratory by cutting cylinders of 3·55 cm diameter and the same height from the samples with non-magnetic, diamond-impregnated tools after the original orientation of the samples had been restored.

3. *Measurements and accuracy of the final results.*—The cylindrical specimens of known orientation were measured on a simple astatic magnetometer. The technique of these measurements has been described elsewhere (**9**, pp. 11–19). The resulting directions were subsequently corrected for the geological dip of the strata, which is respectively 2° E., 2° S.E., 1° S.W. and 1° S.W. for sections A–D. The final directions were thus obtained expressed in declination relative to geographic north and in inclination relative to the horizontal plane, both at the point where the samples were taken.

The uncertainty in the final directions is best discussed under two headings, namely, uncertainty in the declination and uncertainty in the direction in space. The possible error in the declination is 6°, made up of 3° for the uncertainty in the direction of the arrow drawn on the top surface of the sample *in situ*, 1°·5 for the uncertainty in the bearing of the land-mark when read from the topographical map and 1°·5 for the error introduced by transferring the bearing arrow from the top of the sample to the top surface of the cylindrical specimen which is actually measured. The possible error in the direction in space amounts to 9°. It consists of a 2° uncertainty for the marking of the horizontal lines on the sample *in situ*, one of 3° for the restoration of the horizontal position of the sample in the laboratory and its casting in plaster of Paris, one of 1° introduced by the cutting of cylindrical specimens by means of a trepanning tool, one of 2° for the measurements on the astatic magnetometer (for the greater part due to imperfections in the shape or homogeneity of the specimen and to uncertainties in the position of the specimen in its holder) and finally one of 1° in the geological dip of the strata.

Allowing for the fairly steep magnetic dip of the specimens of approximately 60°, the possible error in the declination reduces to only 3° on the sphere. This

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possible error of 3° is now added to the other possible error of 9° in space, even though the former is not strictly speaking an error in space. The total possible error in the final direction of magnetization of a single specimen is therefore 12° in space. In terms of probable errors this amounts to 4°.

4. *Discussion of results.*—In each section a number of lava flows were sampled, consecutively numbered from the bottom to the top of the section. From each flow several samples were taken at intervals of about 4 feet, and from most of these two cylinders were prepared. The number of flows, samples and specimens are shown in Table I. In all, 120 cylindrical specimens were measured.

TABLE I

Section	Number of flows exposed	Number of sampled flows	Label of individual flows	Number of samples from each flow	Number of samples from which two specimens were cut
A Downhill	6	6	A6	3	2
			A5	2	2
			A4	3	2
			A3	2	2
			A2	2	1
			A1	3	3
B Woodtown	9	7	B6	3	2
			B5	3	1
			B4	3	3
			B3	3	2
			B2	3	2
			B1	3	1
			Bo	3	2
C Evish	8	6	C6	4	4
			C5	3	0
			C4	4	3
			C3	3	1
			C2	3	1
			C1	3	3
D Glenariff River	7	5	D5	3	2
			D4	4	3
			D3	3	3
			D2	3	1
			D1	3	2
Total A+B+C+D	30	24		72	48

Flow C6 is the most completely sampled flow and it will therefore be discussed in some detail. In Table II the directions of the natural permanent magnetization of the individual specimens are listed. The north-seeking pole of the natural magnetization of all the specimens points southwards and upwards, i.e. they are reversely magnetized. The same is true of all other lava flows discussed in this paper. In Table II are also listed the angles between pairs of specimens cut from

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the same sample. From the discussion of the possible errors in the preceding section it will be clear that the maximum possible angle between such pairs, consistent with homogeneous magnetization of the sample, amounts to approximately 7°·5. As the maximum angle between the pairs actually observed is 7°, the results are consistent with the idea that the samples are homogeneously magnetized.

TABLE II

Directions of the natural permanent magnetization of the individual specimens from flow C6

Label of specimen	Declination	Inclination	Angle between I and II	Mean directions	
				Declination	Inclination
C6 a I	N. 184°·4 E.	−72°·9	3°	N. 185° E.	−74°
II	191°·0	−74°·5			
C6 b I	192°·0	−63°·8	< 1°	192	−64
II	192°·6	−63°·7			
C6 c I	195°·0	−72°·7	1°	193	−72
II	192°·0	−72°·2			
C6 d I	200°·0	−66°·0	1°	198	−62
II	196°·6	−59°·0			

N.B. C denotes the locality (Evish), 6 is the number of the flow in that section, a–d denote the four samples taken from this flow at intervals of 4 ft. and I and II denote the two cylindrical specimens prepared from each sample. The directions are given before the correction for geological dip, which is 1° S.W.

Similarly the difference in direction between the separate samples (labelled a–d) of flow C6 can be found. The directions of the permanent magnetization of samples C6 a–d are also shown in Table II. The angles between these directions are given below in Table III.

TABLE III

Angles between the directions of magnetization of samples from flow C6

b	c	d	
11°	2°	12°	a
	9°	3°	b
		10°	c

From the discussion of the possible errors in the preceding section it follows that the maximum possible angle between different samples from the same flow consistent with homogeneous magnetization of the sampled part amounts to approximately 20°. The angles in Table III are smaller, thus suggesting that the magnetization is in fact homogeneous.

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We may now enquire how much reliance can be placed upon the mean direction of magnetization of flow C6, calculated from the four directions for samples a–d in Table II. In order to formulate an answer use is made of Fisher's treatment of dispersion on a sphere (**10**). Each direction is represented by a unit vector. The length and direction of the resultant vector (R) is then determined. In Table IV the number of unit vectors (N), the length of the resultant vector (R) and the radii of the circle of confidence (θ) for different probabilities (P) are shown.

TABLE IV

Mean direction of magnetization of lava flow C6:
Declination N. 192° E. Inclination −68°.
Number of unit vectors $=N=4$.
Length of the resultant vector $=R=3{\cdot}981$.
Radius of the circle of confidence: for $P=50$ per cent, $\theta=2°{\cdot}9$.
for $P=5$ per cent, $\theta=7°{\cdot}3$.

This means that the probability is 50 per cent that the true direction of magnetization of the flow differs more than 2°·9 from the calculated mean direction. The same holds for a probability of 5 per cent and an angle of 7°·3. These figures show that the true direction of magnetization can be fairly accurately defined from measurements on a comparatively small number of specimens. For the other flows the situation is less favourable, as fewer specimens are available (*cf.* Table I) but the difference is not large.

The mean directions of magnetization of individual flows have been calculated by finding the mean directions of pairs of specimens and combining these, if necessary, with measurements on single specimens which are not members of a pair. Each direction is given unit weight. If unit weight is given to each specimen, the same mean directions result. The mean directions, expressed in declination and inclination, are given for all flows in Table V.

These directions have not yet been corrected for the geological dip of the strata.

TABLE V

Mean direction of magnetization of individual lava flows before the correction for the geological dip of the strata

Section and geological dip of strata	Flow	Mean directions		Section and geological dip of strata	Flow	Mean directions	
		Decl.	Incl.			Decl.	Incl.
	A6	N. 217° E.	−56°		C6	N. 192° E.	−68°
	A5	175	−60		C5	191	−67
A	A4	208	−73	C	C4	206	−67
2° E.	A3	175	−43	1° S.W.	C3	209	−61
	A2	181	−45		C2	216	−64
	A1	195	−46		C1	224	−65
	B6	219	−60		D5	189	−61
	B5	198	−63		D4	167	−48
B	B4	212	−62	D	D3	162	−46
2° S.E.	B3	186	−42	1° S.W.	D2	185	−44
	B2	176	−75		D1	191	−42
	B1	204	−73				
	B0	203	−71				

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Table V shows that all flows are magnetized in a direction approximately opposite to that of the present field (the north-seeking poles point south and up from the horizontal plane). This phenomenon of natural reverse magnetization has in recent years attracted much attention (*cf.* **12, 4**). One explanation is that it is due to the physical properties of aggregates of ferromagnetic minerals which produce reverse magnetization on cooling in a field which had essentially the same direction as the present field. Other explanations invoke reversals of the local or the main geomagnetic field to account for the observations (for a discussion, *cf.* **4, 9, 12**). As no observations were made on the Northern Ireland plateau basalts which have any bearing on this controversy, this matter is not further discussed here.

In analogy with the results obtained by one of us from a study of Icelandic plateau basalts (**4**) there is every reason to assume that the permanent magnetization of the Northern Ireland plateau basalts originated when these flows cooled down for the first time after their extrusion and that the direction of magnetization has been stable ever since. The mean direction of the natural permanent magnetization, calculated by taking the mean of all directions listed in Table V after each direction has been represented by a vector of unit length, is therefore considered to represent the mean direction of the geomagnetic field in Northern Ireland during the period of time when these plateau basalts were formed or its exact opposite direction. Now it is known from an investigation of postglacial lava flows in south-west Iceland that the mean direction of magnetization of this group of flows agrees better with the direction of the theoretical dipole field than with that of the actual present field. This is shown in Table VI (from **4**).

TABLE VI

Mean direction of natural permanent magnetization of eight postglacial lava flows of south-west Iceland, ranging in age from 2000 *to more than* 6000 *years*

	Declination	Inclination	Angle with mean direction
Mean direction of magnetization	N. 1°·1 E.	+73°·8	0°·0
Present field	N. 27°·0 W.	+77°·0	7°·5
Dipole field	N. 0°	+76°·3	2°·5

The theoretical dipole field is the field due to a magnetic dipole situated at the centre of the Earth and directed along its geographical axis, in other words the field that would result if the magnetic poles coincided with the geographical poles. The declination of such a field is zero, the inclination I can be calculated from the geographical latitude λ by means of the formula $\tan I = 2 \tan \lambda$. The striking agreement between the mean direction of magnetization and the direction of the dipole field leads to the idea that, taken over a period of several thousands of years, the magnetic poles centre on the geographic poles. The period of time covered by the Northern Ireland plateau basalts is difficult to estimate, but the authors are confident that the average time interval between successive flows must be of the order of at least 1 000 years and that consequently a period of at least 10 000 years

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has been covered by the sampled flows. This is sufficient to ensure that the mean direction of magnetization will correspond closely to the dipole field direction.

In Fig. 2 the directions of the natural permanent magnetization of all individual lava flows are shown, after correction for the geological dip. It will be seen that all flows possess reverse magnetization. It appears that there are two groups, a large one of rather steep magnetic dip and a smaller one of less steep magnetic dip. The latter group is formed by flows D1–4, A1–3 and B3. As it is not inconceivable that later local geological processes have changed the original directions of magnetization of some of the flows, the results of Fig. 2 are grouped together in two different ways. The first group includes all directions, the second group includes all directions except those from section D. The latter arrangement serves to test the hypothesis that the samples of section D have been sampled along a fault and may have suffered changes in their directions of magnetization. The results are shown in Table VII. This table shows that the effect of the rejection of the D samples is negligible. We shall therefore exclusively use the first group which includes all directions.

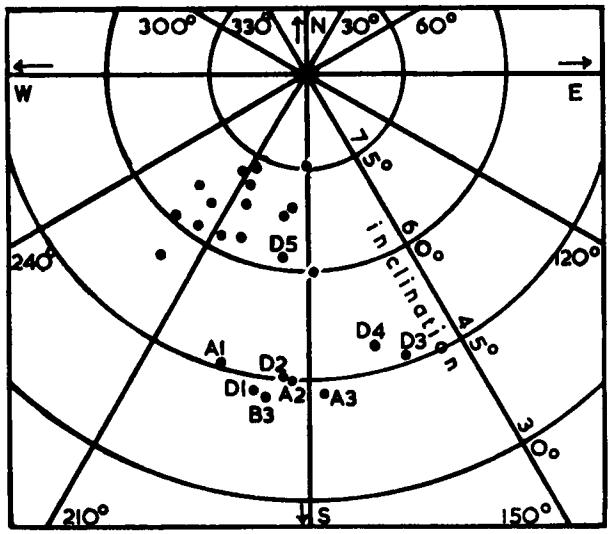


FIG. 2.—*Part of a stereographic polar projection (upper hemisphere) showing the directions of the natural permanent magnetization of all individual lava flows after the correction for the geological dip of the strata has been applied.*

We may simplify the discussion somewhat by taking the opposite direction of the calculated mean direction of magnetization for the purpose of comparison, leaving undecided whether this reverse magnetization is an expression of a reversal of the main geomagnetic field or is due to the magnetic properties of the ore minerals in the rocks. Table VII shows that the agreement of the mean direction of magnetization with the dipole field is better than that with the present field, but that even the former difference is statistically significant, using Fisher's criteria (**10**). We have seen that, taken over a period of several thousands of years, the magnetic poles seem to centre on the geographic poles and, assuming that the same has been true in the geological past, we must conclude that the position of the geographic pole relative to N. Ireland in Eocene time was somewhat different from the present position. Nevertheless, it would be premature to ascribe this difference

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to any such process as continental drift or wandering of the poles. The plateau basalts have been fissured and cut by dykes on an impressive scale (*cf.* **5**, pp. 167–74 and Figs. 60 and 61) and even though the difference between the mean direction of magnetization and the dipole field is statistically significant, the authors prefer to think that local geological processes may be responsible for the slight difference, rather than to conclude from this single set of observations that such controversial processes as continental drift or polar wandering have taken place.

TABLE VII

I. Mean direction of the natural permanent magnetization of all flows

	Declination	Inclination	Angle with mean direction
Mean direction	N. 194° E.	−60°·2	0°
Present field	N. 13° W.	+70°	180°–15°
Dipole field	N. 0°	+70°·8	180°–11°·5

Radius of the circle of confidence (θ) for $P=1$ per cent, $\theta=6°·8$.
$N=24$, $R=23·253$ 5 per cent 5°·4.
10 per cent 4°·7.

II. Mean direction of the natural permanent magnetization of all flows except D1–5

	Declination	Inclination	Angle with mean direction
Mean direction	N. 200° E.	−62°·6	0°
Present field	N. 13° W.	+70°	180°–15°
Dipole field	N. 0°	+70°·8	180°–11°·0

Radius of the circle of confidence (θ) for $P=1$ per cent, $\theta=6°·9$.
$N=19$, $R=18·545$ 5 per cent 5°·4.
10 per cent 4°·7.

The angle between the two mean directions under I and II is 4°·5.

It may be added that when the directions of magnetization of flows D1–5, A1–3 and B3 (i.e. those which form the smaller group of directions of less steep magnetic dip in Fig. 2, plus flow D5) are rejected the mean direction of magnetization has a declination N. 204°·3 E. and an inclination of −66°·3. The opposite of this mean direction forms an angle of 10°·5 with the present dipole field direction. This angular difference is also statistically significant as the radius of the circle of confidence for $P=1$ per cent amounts to 5°·0. The rejection of the above-mentioned part of the evidence therefore does not lead to any better agreement with the present dipole field, nor is it justifiable on physical or geological grounds.

In this connection it is of interest to compare the calculated mean direction of magnetization with the direction of the dipole field in N. Ireland computed for the position of the geographic poles as proposed by advocates of the polar wandering hypothesis. Gutenberg (**13**, p. 202, Fig. 12) has reproduced a diagram by Köppen showing the alleged positions of the north pole from Carboniferous times

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to the present. In Table VIII the position of the north pole in Eocene and Miocene times is shown (from **13**). From these positions can be calculated the alleged direction of the theoretical dipole field in Northern Ireland in Eocene and Miocene times. These directions are also shown in Table VIII and have been compared with the mean direction of magnetization (actually its opposite direction). These directions are shown on a stereographic projection in Fig. 3. Table VIII shows that the angular difference between the mean direction of magnetization and the Eocene dipole field direction is 43°. This discrepancy is so large that the alleged position of the north pole in Eocene times must be considered as incorrect. If the Northern Ireland plateau basalts should appear to be of Miocene age (*cf.* below), the angular difference between the mean direction of magnetization and the (Miocene) dipole field direction is only 14° and the discrepancy is considerably reduced, even though it is still statistically significant.

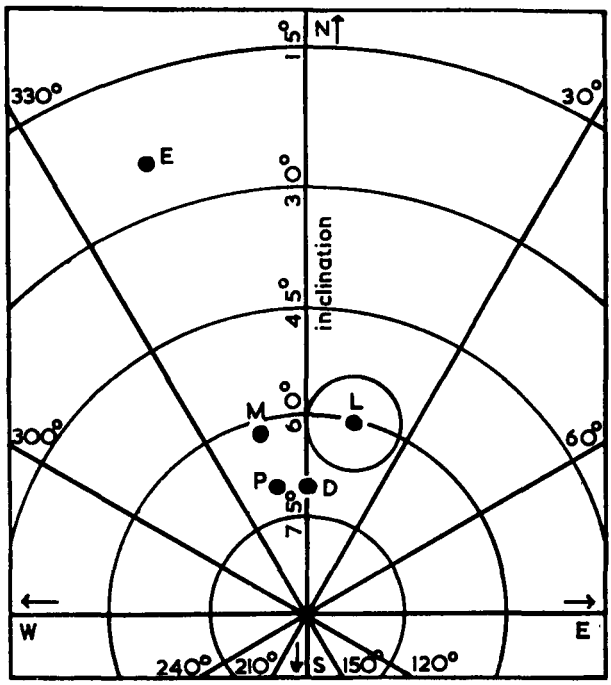


FIG. 3.—*Part of a stereographic polar projection (lower hemisphere) showing the opposite direction of the mean direction of magnetization of the Lower Basalts (L), the direction of the present field (P), the direction of the present theoretical dipole field (D) and the directions of the theoretical dipole fields computed for the alleged position of the geographical north pole in Eocene and Miocene times (E and M respectively). The circle of confidence with radius θ=6°·8 (for probability P=1 per cent) is shown around the mean direction of magnetization.*

It has been stated above that the Irish plateau basalts are of Eocene age. However, there is still some doubt about this. Charlesworth (**5**, p. 144) states that the age of the inter-basaltic horizon which separates the Lower and the Upper Basalts (the Lower Basalts rest on an eroded Chalk surface) is probably Eocene or Oligocene but may be later. Simpson (**14**) concludes from his study of fossil pollen that the vulcanicity in Scotland (Ardnamurchan) did not commence before Early Oligocene times. He considers the inter-basaltic Ulster horizon equivalent to the Mull flora which he has dated as Late Miocene and Early

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Pliocene (personal communication). Should it turn out that a considerably younger age than Eocene has to be assigned to the Lower Basalts of Northern Ireland (e.g. Miocene), then the above argument against the polar wandering hypothesis, as it now stands, will lose its force.

The mean intensity of the natural permanent magnetization of the Lower Basalts is $1{\cdot}7 \times 10^{-3}$ e.m.u.

TABLE VIII

Alleged position of the geographic north pole relative to present-day geographical coordinates:

In Eocene times: 160° *W.*, 45° *N.*
In Miocene times: 150° *W.*, 75° *N.*

Mean geographical coordinates of the sections A–D in N. Ireland:
6°·4 *W.*, 55°·1 *N.*

	Declination	Inclination	Angle with mean direction of magnetization
Mean direction of magnetization	N. 14° E.	+60°·2	0°
Theoretical dipole field in Miocene times	N. 13° W.	+61°·3	14°
Theoretical dipole field in Eocene times	N. 19° W.	+23°·9	43°

Radius of the circle of confidence for $P=$ 1 per cent, $\theta = 6°{\cdot}8$.
10 per cent $4°{\cdot}7$.

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